Abstract

Our study explores the development and comparative evaluation of deep learning models for the automatic detection of fractures in X-ray images in different anatomical areas, such as the hand, leg, hip and shoulder. A complete review of commonly used pre-trained models like ResNet50, VGG16 and Vision Transformer is discussed in conjunction with a series of increasingly complex handcrafted architectures, finishing with the creation of an improved multimodal advanced network that incorporates tabular metadata, such as scan angle, body part and hardware status, along with image features. The evaluation performed using a dataset of 4,083 X-ray images demonstrates that the use of transfer learning from natural image datasets (ImageNet) to medical X-ray images produces significantly effective results, despite the differences between the domains, with pre-trained models far outperforming the handcrafted architectures.

The results reveal that ResNet50 achieved the highest individual performance (89.53% accuracy, 0.9313 ROC-AUC), while an ensemble combining ResNet50 and Vision Transformer further improved results (89.03% accuracy, 0.9347 ROC-AUC). Despite architectural innovations in our handcrafted models, including regularization techniques and multimodal fusion of tabular data, the performance gap between our best handcrafted model (78.82% accuracy) and pre-trained models exceeded 10 percentage points, underscoring the considerable advantage of transfer learning from large-scale datasets. These findings contribute to the growing evidence for the efficacy of deep learning in fracture detection and highlight the value of ensemble methods and ImageNet derived features despite substantial domain differences between natural images and radiographs.

1 Introduction

1.1 Background and motivation

Identifying fractures in radiographs represents an important and essential diagnostic task in the practice of emergency medicine, orthopedic practice and radiology. Every year, countless numbers of X-ray examinations are performed globally for the examination of potential fractures that may have arisen from injuries such as trauma, falls, sport injuries and other pathological conditions [1]. Accurate and timely diagnosis of fractures is highly significant in the administration of appropriate patient treatment because undiagnosed fractures lead to improper recovery, prolonged pain, loss of function and increased healthcare spending.

Fracture detection represents a core practice within the field of radiology. However, many fractures are expressed as subtle disruptions within the bone's continuity and are therefore difficult to detect on conventional radiographs. Technical factors such as patient positioning, exposure parameters and post-processing filters also have the potential to significantly affect fracture characteristics' detectability. Radiologists and emergency physicians consecutively analyze these images within significant patient volumes and high clinic pressures, consequently increasing the potential for perceptual failures.

These challenges lead to the troubling frequency of missed fractures in practice, as shown by the evidence that radiologist interpreted errors have rates between 1% and 4% and even higher rates for non-radiologist physicians [2, 3]. Missed diagnosis of these conditions has the potential to result in serious harm to patients and accounts for significant numbers among medical malpractice claims [4].

The advent of deep learning (DL) techniques, specifically through convolutional neural networks (CNNs) and, more recently, transformer models, has shown great promise in addressing these challenges. These techniques provide consistent performance metrics that are not influenced by factors like fatigue, workload, or time pressure. DL excels at detecting complex patterns that may be missed by human observers, thus enabling the detection of often-missed fractures. In addition, artificial intelligence (AI) solutions can rank cases with a high likelihood of fracture, thus improving the operational efficiency of radiologists and reducing the time taken for diagnosing critical results. Moreover, AI can serve as an auxiliary "second opinion," offering an additional level of verification that complements human evaluation and may reduce interpretative errors. Finally, learning systems have the ability to highlight subtle anomalies, which could accelerate the educational progress of both radiology residents and general practitioners.

The integration of AI assistance in fracture detection aligns with broader healthcare goals of improving diagnostic accuracy, standardizing care and optimizing resource utilization. As imaging volumes continue to increase globally while radiologist shortages persist in many regions, computational assistance becomes increasingly valuable for maintaining quality care. Moreover, the potential for AI to democratize expertise by bringing specialist-level diagnostic capabilities to underserved areas represents a compelling motivation for developing robust fracture detection algorithms.

1.2 Problem statement and objectives

The primary question we examine relates to the development and evaluation of DL models for binary classification of bone fractures using X-ray images. Our study focuses particularly on the distinction between fracture-positive and fracture-negative X-rays in diverse anatomical regions, such as the hand, leg, hip and shoulder, a task that presents unique difficulties.

Unlike many previous investigations that restrict their scope to singular anatomical regions (e.g., the wrist alone), our study deals with the more intricate problem of fracture detection across a variety of anatomical structures with varying bone densities, configurations and overlapping patterns. Binary classification essentially answers the clinical question (Is there a fracture?) that guides initial management decisions and determines the need for orthopedic treatment. To be clinically relevant, such models must demonstrate generalizability to various fracture types, patient populations and imaging modalities.

Furthermore, although accuracy optimization is paramount, practical model application requires a delicate balance between performance and computational efficiency. Our study seeks to evaluate and contrast the accuracy of a diverse set of DL models, including both established pretrained architectures (ResNet50, VGG16 and Vision Transformer (ViT)) and a suite of handcrafted custom-built networks. These self-developed models span a range from a simple convolutional structure to deeper, regularized architectures and even advanced configurations that integrate multimodal inputs, thereby complementing the transfer learning approaches derived from natural image datasets. Additionally, the study prioritizes the development of an optimal preprocessing and augmentation strategy tailored to the distinct characteristics found in radiographic X-ray images. Our research further investigates ensemble methods that

combine the complementary strengths of each model to enhance fracture detection outcomes. Finally, the study examines how hyperparameter selection, training methodologies and regularization strategies affect performance within this specialized medical classification task.

Within this work, we recognize the extensive clinical relevance of transfer learning by employing models pretrained on ImageNet, even in the challenging context of bridging the gap between natural images and X-ray data. Our research extends this approach by integrating custom-architected models specifically devised for X-ray analysis.

1.3 Novelty and contributions

Our mini-project has multiple key contributions to the area of fracture detection through DL. Unlike the majority of the current studies that address a single anatomical zone, our approach addresses the far more difficult problem of fracture detection from diverse anatomical structures (hand, leg, hip and shoulder). This multi-anatomical approach requires models to learn features broad enough to identify fractures under very much varying bone density, structure and overlapping patterns. Secondly, we offer a detailed comparative study of popular pre-trained models and an incremental sequence of handcrafted designed networks explicitly tailored for X-ray analysis, with interesting conclusions regarding the transferability of features from natural images to the medical domain.

One of the key contributions of our work is the development of a multimodal approach that uses tabular metadata: scanning orientation, body location and hardware status, alongside image features as inputs to our advanced model architecture. This is a departure from the conventional image-only approaches and allows the model to leverage contextual information potentially influencing fracture appearance and detection. In addition, our work evaluates ensemble approaches that aggregate the respective strengths of architectures with different inductive biases and demonstrates performance improvements over individual models.

Through the search for model architectures, training procedures and regularization methods, our work provides evidence of the merits of transfer learning and pre-trained models for fracture detection compared to handcrafted models. The generalizability of our pre-trained models, in the presence of domain discrepancy between natural images (from ImageNet) and X-rays, provides compelling evidence to the argument on the transferability of ImageNet acquired features to specialized medical imaging tasks. The findings have both theoretical implications on the explanation of cross-domain knowledge transfer and practical implications on the construction of clinical decision support systems.

2 Related work

The recent study by Alshahrani and Alsairafi [5] discusses a system for bone fracture classification based on the use of convolutional neural networks. In this work, the authors focus on training the YOLOv8 model to enable real-time object detection and image segmentation, in addition to using the VGG16 architecture for fracture classification, using the FracAtlas dataset containing 4,083 X-ray images. By employing thorough hyperparameter tuning and data augmentation strategies, their approach achieved a test accuracy of 72.22% using the VGG16 model. The results of their work indicate the high potential of classical CNN models, especially when subjected to extensive optimization, to aid clinical diagnostics by increasing the accuracy and speed of fracture detection.

On the other hand, Ruhi et al. [6] introduced a transfer learning approach using attention mechanisms along with pre-trained networks like InceptionV3 and DenseNet121 with the objective of improving the accuracy of bone fracture classification. Their extensive experimental studies, which include comparative analyses of pre-trained and non-pre-trained models, establish that transfer learning-based models are capable of achieving outstanding accuracy of over 90%. This paper not only highlights the central role played by state-of-the-art deep learning methods in the area of medical imaging but also highlights the major role of transfer learning in promoting the development of more accurate fracture detection systems.

In another study, Schilcher et al. [7] developed a multi-modal neural network which considers both, X-ray images and additional health records (clinical tabular data) as inputs to detect atypical femur fractures (AFF), a specific but a very rare type of fracture. They process the x-ray image data through a pre-trained model, ResNet50 and extract the image features and on a separate network branch they process the clinical tabular data. These branches (modalities) are then fused and the model provides us with a prediction. It was observed that integrating the clinical tabular data along with the x-ray image data yielded some improvements in the model's diagnostic performance. The ROC-AUC increased from 0.966 when only X-ray image data was used to 0.987 (even a little increase goes a long way in

diagnostic applications) and sensitivity improved significantly from 0.796 to 0.903. This demonstrates the potential of also considering additional medical information along with X-ray image data via a multi-modal approach to enhance the model's diagnostic performance.

3 Dataset and data preprocessing

3.1 Dataset description and visualization

The dataset used in this project is publicly available on Kaggle [8] and it consists of a total of 4083 x-ray images and one csv file containing additional information related to each x-ray image. Out of the 4083 images, 3366 images are labelled as non-fractured and the remaining 717 images are labelled as fractured. Although all the images have a JPEG format with three bands, they have varying sizes. These x-ray images are taken from various sections of the human body and in various angles. The fractured images could have a single or multiple fractures. The images, also could have foreign material such as pins, screws, plates, etc. All these additional information is mentioned in the csv file.

More specifically, the csv file mentions which body part the x-ray is taken from using binary variables, one for each body part (hand, leg, hip and shoulder) and if it is taken from multiple body parts, then more than one of these binary variable is set to 1 along with another binary variable named `mixed` which is set to 1. The file also mentions the angle at which the x-ray is taken from using binary variables, one for each angle (frontal, lateral and oblique) and if the x-ray consists of multiple scan angles, then more than one of these binary variables is set to 1 along with another binary variable named `multiscan` which is also set to 1. Penultimately, the file also mentions if the x-ray image has any foreign particles such as (pins, screws, plates, etc.) using a binary variable named `hardware` which is set to one indicating presence of such foreign particles. Lastly, the file mentions whether or not the x-ray image is fractured or not using a binary variable fracture which is set to 1 in case fracture(s) exists and also mentions the number of fractures using an integer variable `tfracture_count`. All these variables can be checked in Appendix A.

Figure 5 shows some example images of our dataset, with the nature and variability of fracture and non-fracture examples.

3.2 Data preparation and augmentation

3.2.1 Handcrafted models

The additional data from the csv file is represented in a poor manner. Hence the first step was to represent the data in a more optimal manner. In order to do this, the individual binary variables were converted into categorical variables.

More specifically, the binary variables providing information regarding the body part from which the x-ray image was taken `namely`, `hand`, `leg`, `hip`, `shoulder` and `mixed` was replaced by one categorical variable `body_part`. This variable mapped the different combinations of body parts into a single integer value where each integer value represents a unique combination of the body parts. Similarly, the binary variables providing information regarding the angles from which the x-ray was scanned namely, `frontal`, `lateral` and `oblique` was replaced by one categorical variable `angles`. This variable mapped the different combinations of scan angles into a single integer value where each integer represents a unique combination of the scan angles. The variables `hardware` and `fracture` retained its structure. Lastly, the variable `fracture_count` was removed as the focus of this project was to detect the presence of a fracture rather than estimating the number of fractures. The mapping of the new csv variables can be check in Appendix B.

The x-ray images are read and are first converted into RGB bands from BGR bands. Then these images are greyscaled as the focus during detecting fractures is on the structural image features of the image and not on colour. Lastly, these images are re-sized to 224 x 224 pixels in order to ensure uniformity.

Once the aforementioned transformations are made, the data is split into training-validation data and testing data with a ratio of 80:20. This split is made in a stratified manner in order to account for the class imbalance. The training-validation data is then used to create 5 stratified folds with each fold containing its own training and validation subsets for cross-validation.

The 5 training subsets, 5 validation subsets and testing subset is then converted into 11 separate `tfrecord` files in which each record example contains the variables `raw_image` (raw image data post transformation), `body_part`, `angles`, `hardware` and `fracture`. This conversion is performed using parallel processing to speed up the process and to ensure efficient data loading and storage.

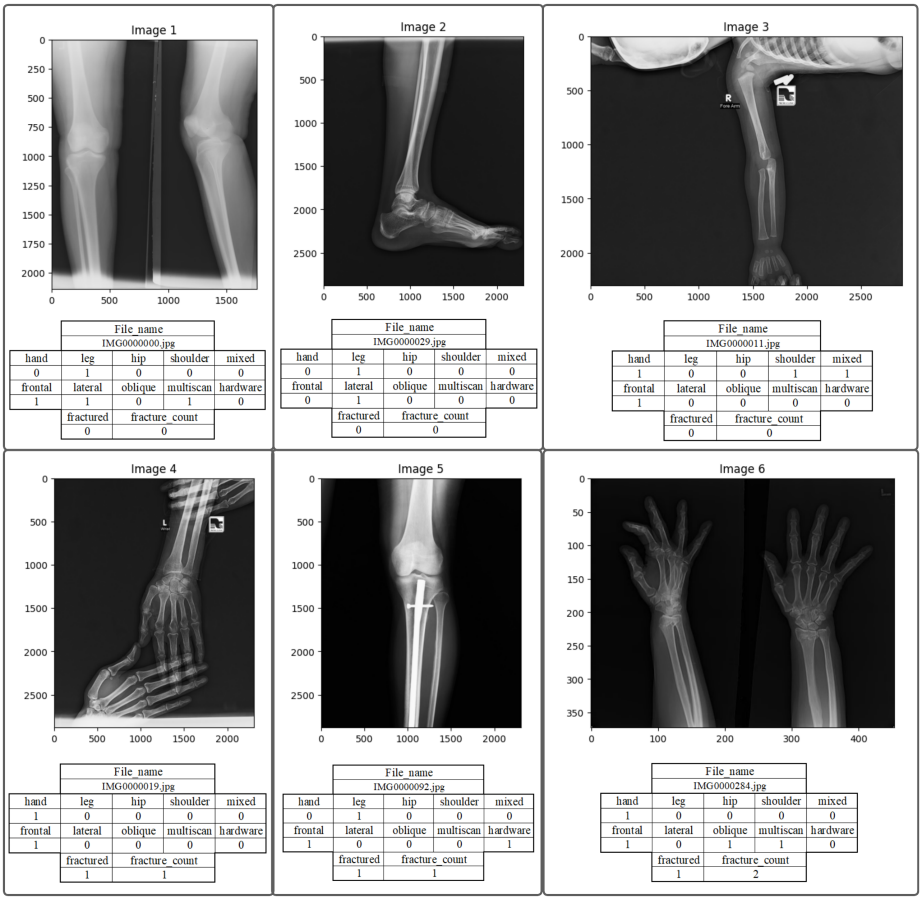

Figure 1: Examples of fractured and non-fractured images along with their additional information.

In order to create the 5 training datasets, 5 validation datasets and 1 testing dataset, which will be used further, the data in the `tfrecord` files is then read in a shuffled manner with pre-fetching enabled for efficient loading. The x-ray images from the 5 training subsets are augmented using techniques such as random horizontal and vertical flipping and random rotation (90°, 180°, 270°) with equal probability. A dictionary named `features` is then created, containing the variables `raw_image` and `tabular_input`. The `tabular_input` in-turn is a dictionary containing the variables `body_part`, `angles` and `hardware` that we stored. The target variable `fractured` is stored separately in a variable called `label` in order to follow the structure accepted by the `TensorFlow` module.

3.2.2 Pre-trained models

The pre-trained models only use the x-ray image data and not the additional information contained in the csv file. The data used in the pre-trained models undergo data preparation and augmentation differently. For pretrained models, data preparation follows a more standardized approach to align with the training protocols of these architectures. Since these models were originally trained on the ImageNet [9] dataset, dataset, we adopted preprocessing techniques that are consistent with their original training methodologies. All images were normalized using the mean and standard deviation values from ImageNet ($\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$) to match the input distribution expected by the pretrained model, allowing the models to leverage their pretrained weights effectively.

To standardize the labelling and ensure consistency across all models, we explicitly remapped the class labels. The default behavior of *PyTorch*'s `ImageFolder` is to assign labels alphabetically, which would have assigned label 0 to "Fractured" and label 1 to "Non fractured". To improve interpretability, we reassigned the labels so that class 1 represents "Fractured" (positive class) and class 0 represents "Non-fractured" (negative class).

For pre-trained models, the data was split into the training set, validation set and testing set with a ratio of 70:15:15 in a stratified manner to handle the class-imbalance and the data loaders for these sets was

configured with a batch size of 32, balancing computational efficiency and gradient stability. Moreover, the training sets were loaded in a shuffled manner to enhance randomization, whereas validation and test sets were loaded as is for reproducibility.

The input resolution was standardized to 224×224 pixels, which is the standard input size for most ImageNet-trained architectures. For validation and testing, we used a consistent approach of resizing to 256 pixels on the shortest edge followed by a center crop to 224×224 pixels. This approach preserves the aspect ratio during resizing and then crops the center portion to obtain a consistent input size.

In order to enhance model robustness and prevent overfitting, we implemented an extensive augmentation pipeline for the training set. We used augmentation techniques such as random cropping where we crop a portion of the image and resize it back to 224×224 pixels, random horizontal and vertical flipping, random rotation up to 10 degrees (to simulate minor variations in positioning), random jittering where we vary the brightness, contrast, saturation and hue with factors of 0.2, 0.2 and 0.1 respectively and random erasing where we erase regions of the image forcing the model to rely on global features rather than specific local patterns that might not generalize well, all with equal probability. This augmentation strategy is more aggressive than what we use for handcrafted models because pretrained networks have stronger regularization capabilities due to their depth and the diversity of features they've learned from ImageNet. The aggressive augmentation helps prevent overfitting and improves generalization, especially given the limited size of our data.

We used minimum changes for testing and verification, only those that offer correct sizing and normalization, such as resizing to 256 pixels on the shortest side keeping the aspect ratio, center cropping to a size of 224×224, transforming them to a *Pytorch* tensor and apply ImageNet's normalization.

For all the pre-trained models we used the initialization approach known as transfer learning using pre-trained weights from ImageNet for the backbone while replacing the final classification layer with a randomly initialized linear layer suited for our binary classification task. This approach leverages the rich feature hierarchies learned from natural images while allowing task specific adaptation. Although medical X-ray images differ significantly from ImageNet's natural images, the low-level features (edges, textures) and mid-level patterns remain useful, particularly when fine-tuned on our dataset

4 Methodology

4.1 Custom CNN architectures / Handcrafted models

We initially developed a simple convolutional model (simple model) to establish a baseline results. Then we added to the model complexity by adding deep convolutional layers creating a deeper convolutional model (deep model). Then to ensure that the deep model learns general features and not specific local features from the training data we added regularization techniques creating a regularized deep convolutional model (regularized deep model). Then finally, we extended the regularized deep model by flattening its output and concatenating it with the outputs of a fully connected neural network which incorporates the additional (tabular) data, allowing the model (advanced model) to leverage both the x-ray image data and also the additional data.

4.1.1 Simple model

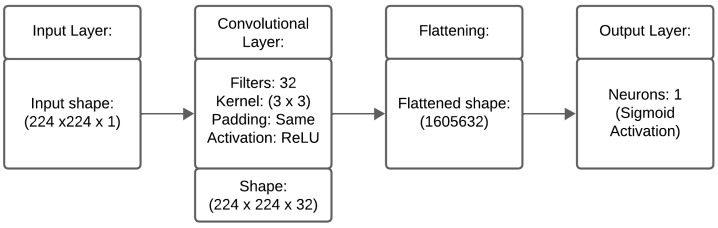

Figure 2: Simple Model Structure.

The simple model is a minimal convolutional neural network (CNN) designed to provide us with baseline results that we could compare our more complex with. It consists of a single convolutional layer which is then flattened and lastly connected to a dense output layer which provides us with binary classifications.

More specifically, the model takes grayscale x-ray images having shape $224 \times 224 \times 1$ as input. This input layer is then connected to convolutional layer with 32 filters having a kernel size of 3×3 with no padding and a ReLU activation. The output of this layer is then flattened and passed to a fully connected (dense) layer with a single neuron and a sigmoid activation function. The outcome is a probability of how likely the model thinks a fracture exists.

4.1.2 Deep model

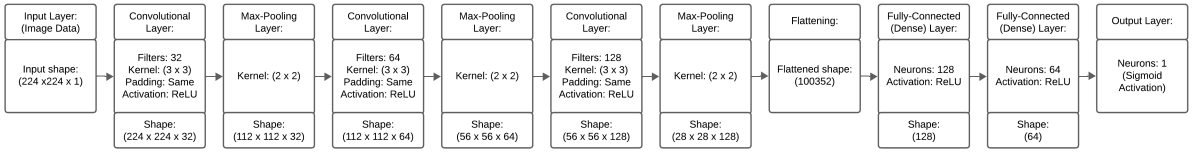

Figure 3: Deep Model Structure.

After establishing the baseline results using the simple model, we aimed to create a model that has an increased capacity to learn complex features and patterns from the x-ray images. To achieve this, we designed a deep convolutional neural network by introducing multiple convolutional and fully connected layers.

More specifically, the model takes grayscale x-ray images having shape $224 \times 224 \times 1$ as input. This input layer is then successively connected to three convolutional layers each followed by a max pooling layer to progressively extract key features from the x-ray images. The three convolutional layers have 32, 64 and 128 filters respectively while having the same kernel size of 3×3, no padding and ReLU activation. The max pooling layer has a kernel size of 2×2 and is performed to downsample the feature maps, promoting spatial invariance by reducing the model's sensitivity to the precise location of features.

After the convolutional and max-pooling layers, the extracted feature maps are flattened and is then passed through 3 fully connected (dense) layers. The first and second layer consists of 128 neurons and 64 neurons respectively and have a ReLU activation. This is followed by another fully connected (dense) layer with a single neuron and a sigmoid activation function which outputs a probability indicating the likelihood of a fracture.

4.1.3 Regularized deep model

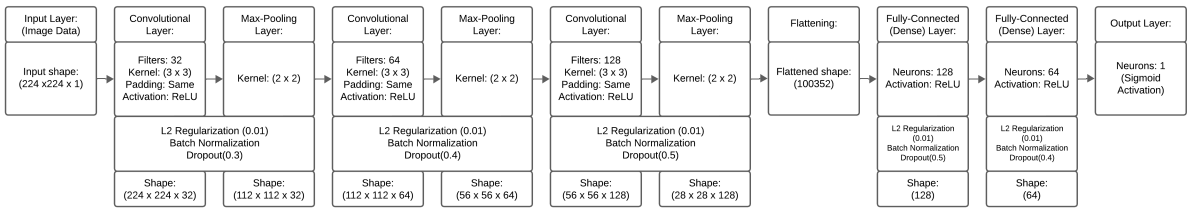

Figure 4: Regularized Model Structure.

During the training of the deep model, it was observed that the loss significantly decreased as compared to the simple model, indicating a massive improvement in learning but this significant reduction did not correspond equally with other metrics such as accuracy, ROC AUC, etc., which hinted towards the model overfitting to the training data. In order to address this issue, we incorporated regularization techniques in the regularized deep model to help the model generalize better, prevent overfitting and provide better results on un-seen data.

The regularized deep model retained the same structure as that of the deep model but incorporated regularization techniques. Every single convolutional layer and fully connected layer underwent l2 regularization with a factor of 0.1, encouraging the model to learn simpler and more generalizable features. Additionally, every layer also underwent batch normalization, which provided stabilization and improved the convergence. Lastly, the three convolutional layers underwent dropout with a rate of 0.3, 0.4 and 0.5 respectively and the first two fully connected layers underwent dropout with a rate of 0.5 and 0.4 respectively. This randomly dropped neurons and set weights to 0, preventing overfitting and forcing the model to not rely on specific complex patterns but on more robust and generalized ones.

4.1.4 Advanced model

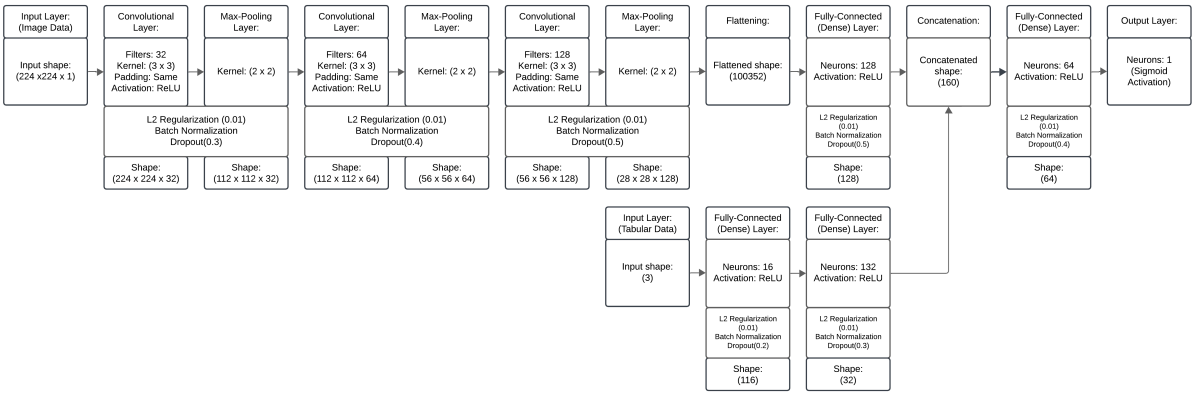

Figure 5: Avanced Deep Model Structure.

The advanced model builds upon the regularized deep model by also considering additional (tabular) information along with the x-ray image data. The image network follows the same structure as the regularized deep model until the second last layer processing the grayscale x-ray images through three regularized convolutional and max-pooling layers followed by flattening the output. The flattened output is then passed to a fully connected layer with 128 neurons and ReLU activation (the second last layer of the regularized deep model) where this branch culminates.

The model also processes tabular data comprising the 3 aforementioned variables, `body_part`, `angles` and `hardware` through a separate network branch. This branch consists of two fully connected layers with 16 and 32 neurons respectively with ReLU activation, wherein each of the layers are regularized in the same manner as explained before with dropout rates of 0.2 and 0.3 respectively.

The outputs of the image network (128 neurons) and the tabular network (32 neurons) are then concatenated. This is then passed through another regularized (in the same manner with a dropout rate of 0.4) fully connected layer with 64 neurons and ReLU activation (the last layer of the regularized deep model) and is then finally passed to another fully connected layer with a single neuron and a sigmoid activation function which outputs a probability indicating the likelihood of a fracture.

4.2 Pre-trained models and transfer learning

Transfer learning has become a powerful methodology in medical image analysis, particularly in situations in which there is limited availability of training images. Here we used three state-of-the-art architectures pretrained on the ImageNet dataset: ResNet50, VGG16 and ViT. Each of them brings different attributes to fracture classification.

4.2.1 ResNet50

ResNet50 [10] is a 50-layer deep residual network that alleviates the problem of vanishing gradients by introducing skip connections. This design enables the effective training of very deep neural networks. ResNet50 has five different phases, each having multiple residual blocks. Every residual block has a "shortcut connection" that skips two or three layers, which are made up of convolutions, batch normalization and ReLU activation functions. In our configuration, we preserved the original architecture while replacing the last fully connected layer. The original classifier mapped 2048-dimensional features to 1000 ImageNet classes, while our modified classifier maps these 2048-dimensional features to 2 classes (fractured vs. non-fractured).

We performed fine-tuning on the entire network rather than limiting it to only the final layers, allowing the model to adapt its feature extraction mechanisms to the unique characteristics of x-ray images. This full fine-tuning approach was chosen because x-ray images are quite different from natural images in terms of their texture, contrast levels and semantic properties (fractures tend to be expressed by detailed line patterns that are poorly captured by the features learned from ImageNet) and our dataset was large enough to allow for successful fine-tuning without causing major overfitting problems.

ResNet50 offered numerous benefits for our task but also presented some challenges. Among its advantages were a deep structure capable of learning intricate details at various levels, residual connections

that mitigate the vanishing gradient problem and enable smoother information flow, well-established performance across diverse computer vision tasks and relatively modest computational requirements compared to larger models. On the other hand, challenges included the risk of overfitting when using small medical datasets, increased complexity which makes the network harder to interpret than simpler architectures and a tendency to focus on texture patterns rather than the structural features critical to identifying fractures.

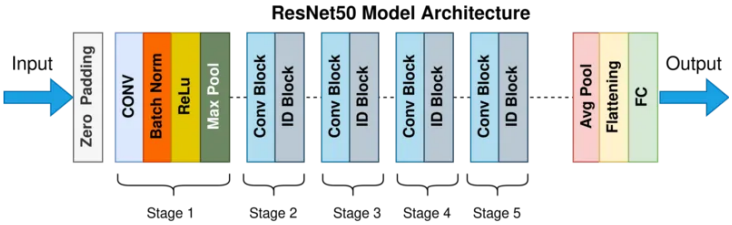

Figure 6: ResNet50 architecture diagram [11].

4.2.2 VGG16

VGG16 [12] is a 16-layer neural network known for its straightforward design, employing 3 x 3 filters for convolution throughout. VGG16 consists of five groups of convolutional layers followed by three fully connected layers. For our particular case, we retained the initial five convolutional blocks and max-pooling layers as the feature extractor, while modifying the classifier. Specifically, we left the first two fully connected layers (each with 4096 units) unchanged and replaced the last layer as the original classifier maps 4096 features to 1000 ImageNet classes, whereas our modified classifier maps these features to 2 classes (fractured vs. non-fractured).

Similar to ResNet50, we fine-tuned the entire network, including both the convolutional and fully connected units. However, because VGG16 is a less complex network that relies heavily on parameters in the fully connected layers, we encountered a greater risk of overfitting, slower convergence (especially during the early training epochs) and heightened sensitivity to learning rate settings.

VGG16 thus possessed both strengths and weaknesses. Its advantages include an easy-to-understand design that is effective for feature extraction, interpretability of its feature maps compared to deeper models and proven proficiency in transfer learning for edge and texture detection. The challenges, however, include an increased likelihood of overfitting due to the high number of parameters in the fully connected layers, higher memory requirements during training, less effective use of parameters compared to more recent architectures and a limited receptive field resulting from the small filter sizes.

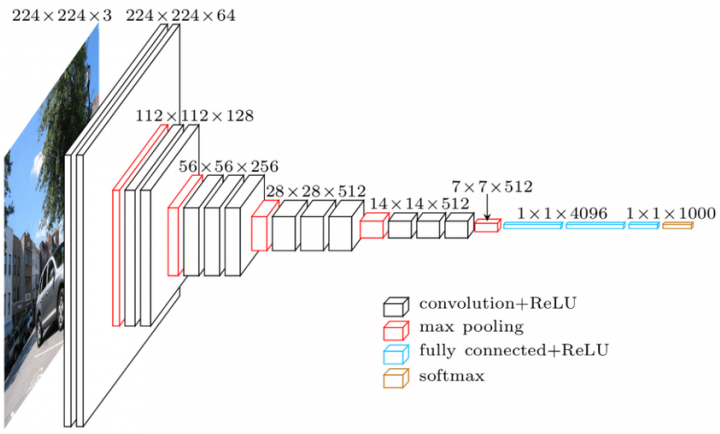

Figure 7: VGG16 architecture diagram [13].

4.2.3 Vision Transformer

ViT [14] represents a paradigm shift in computer vision by applying the self-attention mechanism, originally developed for natural language processing, to image classification tasks. The variant we employed,

ViTB/16, processes images by partitioning them into patches of 16 x 16 pixels, linearly embedding each patch into a token, adding positional embeddings to maintain spatial information and then processing the sequence of tokens through 12 transformer encoder blocks.

In our implementation, only the last classification layer was replaced. Originally, the classification layer converted 768-dimensional features to 1000 ImageNet classes but in our modified version, it converts the 768-dimensional features to 2 classes (fractured vs. non-fractured).

Fine-tuning ViT required special considerations. We used a smaller learning rate (1×10^{-4}) to avoid compromising the pretrained attention mechanisms. Despite its large parameter count, ViT exhibited robust regularization properties. However, the model demanded more computational power and a longer training time compared to CNN-based models.

ViT offered distinct tradeoffs. The advantages include the self-attention mechanism's capacity to detect long-range dependencies within an image, a decreased inductive bias relative to local structure (which enables better adaptation to X-ray characteristics), high scalability for larger datasets and models and an aptitude for capturing the global context that is important for identifying complex patterns of disruption. Conversely, challenges involve increased computational load during both training and inference, reduced efficacy on limited training sets, the absence of a clearly defined hierarchical framework that can impede feature learning and a less established presence in medical imaging applications compared to traditional CNNs.

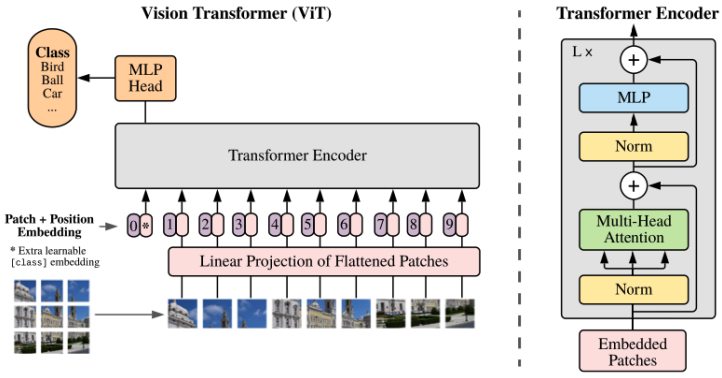

Figure 8: ViT architecture diagram [14].

4.2.4 Ensemble methods

In order to leverage the strengths of multiple architectures complementing each other, we ventured into ensemble methodologies. Specifically, we used soft voting where the output probabilities from the different models were combined before the final classification decision. Additionally, we tested the combinations of multiple models in order to identify the most optimal ensemble. Through our experiments with different ensemble combinations, we found that the ensemble of ResNet50 and ViT yielded the best results.

4.3 Training protocol

The training protocol used to train the handcrafted models was quite similar as compared to the training protocol used to train the pre-trained models. The training protocol used for the pre-trained models was comparatively more tailored and refined as compared to that of the handcrafted models. The details and specifications are mentioned bellow.

As the data possessed a significant class-imbalance, we had to use an appropriate loss function, namely, cross-entropy loss function with class weighting.

For a particular sample i, the loss function is defined as,

$$L_i(y_i, \hat{y}_i) = -\Big(w_1\, y_i\, \log(\hat{y}_i) + w_0\,(1 - y_i)\, \log(1 - \hat{y}_i)\Big),$$
$$w_1 = \frac{N}{N_1}, \qquad w_0 = \frac{N}{N_0}.$$

Where:

- $y_i \in \{0, 1\}$ is the true class label of the sample.
- $\hat{y}_i \in (0, 1)$ is the predicted probability of the positive class. (If $\hat{y}_i$ is exactly 0 or 1, a small value ε is subtracted or added, respectively, for numerical stability.)
- N_1, N_0 and N denote the number of samples in class 1, class 0 and the total number of samples, respectively.
- w_1 and w_0 are the class weights used to compensate for the class imbalance (they are inversely proportional to the class frequencies).

This weighting approach increases the loss contribution of the minority class (fractures), encouraging the model to pay closer attention to such instances and counteracting the model's inclination to predict the majority class.

For the pre-trained models, the Adam optimizer was used with a learning rate of 0.0001, selected through preliminary experiments to balance convergence speed with training stability, the default Adam weight decay, as we found class weighting and data augmentation provided sufficient regularization and a batch size of 32 to optimize GPU memory utilization while providing stable gradient estimates. A stepwise learning rate schedule was also used. The initial learning rate was set to 0.0001, with a step size of every 5 epochs and a gamma of 0.1 (representing a reduction of learning by 90% for each iteration). This schedule supports quick convergence in the earliest epochs and then enables refined changes as training advances, helping to avoid oscillations around local minima in later stages. In contrast, for the handcrafted models, we used the default settings of the Adam optimizer.

For both, pre-trained models and handcrafted models the maximum number of epochs was set at 100. However, in order to prevent overfitting, we implemented early stopping based on validation ROC-AUC (Receiver Operating Characteristic, Area Under the Curve) as it better indicates discriminative ability than accuracy, especially with imbalanced classes. The patience was defined as 10 epochs, giving enough time for the model to recover from temporary performance plateaus. Consequently, training typically ended earlier than 100 epochs, with the best model as determined by the validation ROC AUC was saved for final evaluation

For pre-trained models, we leveraged several techniques to accelerate training, including mixed-precision training where, with access to a GPU, we used automatic mixed precision (AMP) to increase computational efficiency without compromising numerical accuracy and multi-worker data loading, utilizing 2 workers for efficient data preprocessing and loading to minimize CPU bottlenecks.

The handcrafted models were trained on the T4 GPU available on Google Colab which performed the required computations with sufficient hardware acceleration. On the other hand for the pre-trained models training was conducted on an NVIDIA L4 GPU located in the Minerva cluster at Chalmers (made available to us from another course). It used CUDA support where possible, falling back onto Apple MPS or the CPU on our laptops where needed. Lastly, in order to ensure reproducible results, we applied random seeds wherever possible.

5 Results and discussion

5.1 Training and validation loss

In order to analyse the learning behaviour of the models, we tracked the training and validation loss over the epochs. Each of the handcrafted models (Simple, Deep, Regularized Deep and Advanced models) and pre-trained models (ResNet50, ViT, VGG16) were trained separately and their trends are presented in Figure 9.

The striking feature of the invariably lower validation loss compared to training loss in most models is interesting and can be attributed to several factors. The use of regular (in handcrafted models) and aggressive (in pre-trained models) data augmentation methods across all training stages produces more challenging examples for the model to classify, regularization techniques, including dropout and random erasing (in pre-trained models) are applied during training but not during validation. This effect demonstrates that our augmentation method successfully prevents overfitting and enhances the ability of the models to generalize to unseen data.

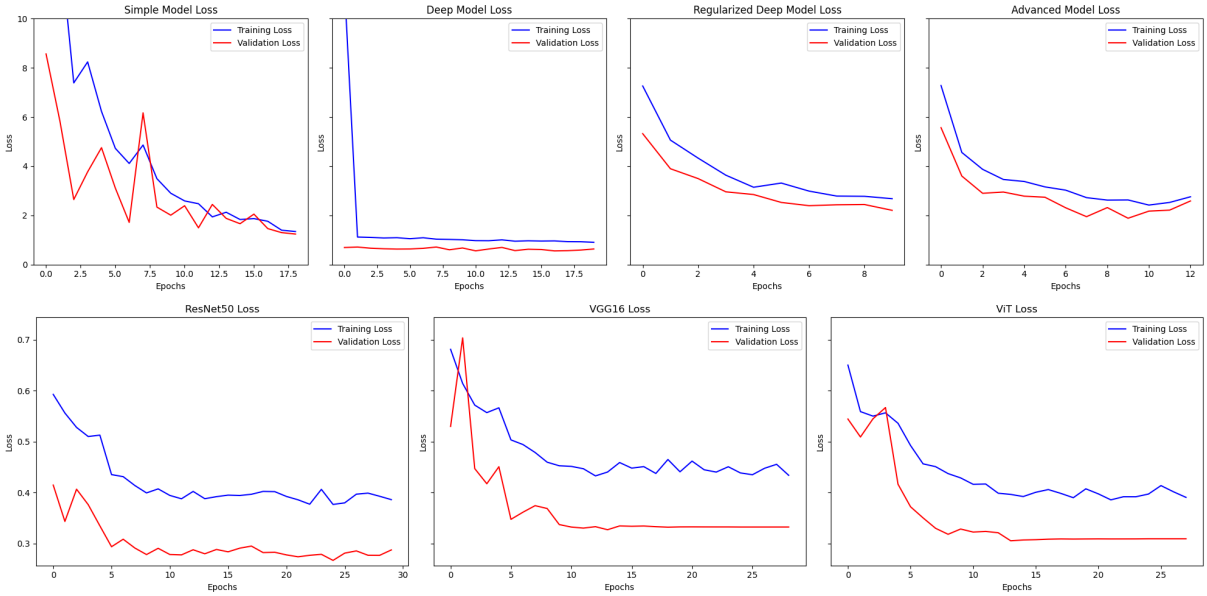

Figure 9: Training and validation loss curves for handcrafted and pre-trained models over the training process.

5.2 ROC curves and AUC analysis

For all models, we plotted the ROC-AUC (Figure 10), which shows the balance between the true positive rate or sensitivity and the false positive rate or 1-specificity as the classification threshold varies.

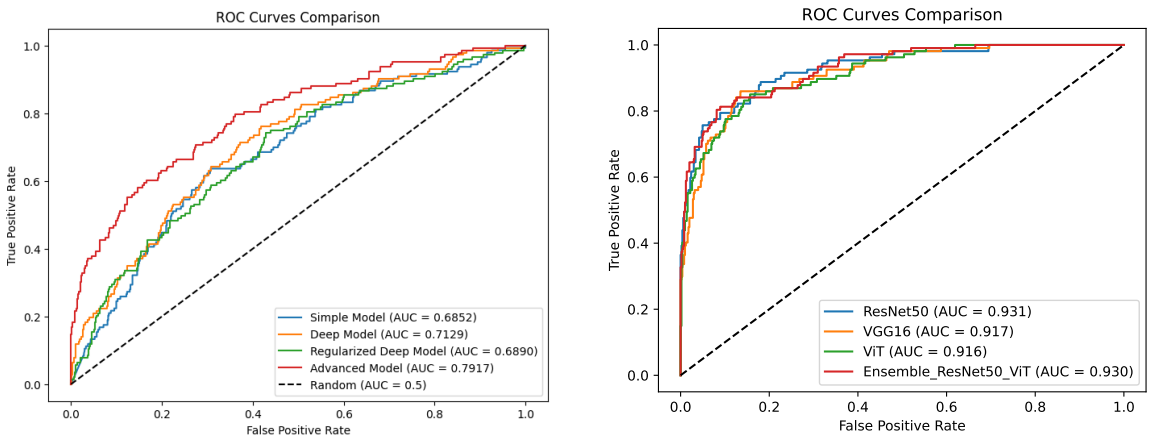

Figure 10: ROC-AUC curves for all models (including ensemble methods).

5.3 Evaluation metrics

We evaluated every model over test data using various evaluation metrics such as accuracy, precision, recall (sensitivity), specificity, negative predictive value (NPV), F1 score and ROC – AUC.

5.4 Comparative analysis

ResNet50 performed better than VGG16, which is typical in computer vision literature. ResNet's skip connections and fewer parameters (25M vs 138M) likely contributed to better gradient flow and generalization to our data. ViT's performance confirms that its self-attention mechanism is effectively capturing long-range dependencies in X-ray images even without CNN-style inductive biases. The best performance of the ensemble confirms that architectural diversity leads to synergistic gains when combined.

Table 1: Performance metrics for handcrafted and pre-trained models (including ensemble).

Model	Accuracy	Precision	Recall	Specificity	NPV	F1	ROC-AUC
Handcrafted models							
Simple Model	0.6095	0.2596	0.6643	0.5979	0.8936	0.3733	0.6851
Deep Model	0.6940	0.3026	0.5734	0.7196	0.8883	0.3961	0.7129
Regularized Deep Model	0.7687	0.3380	0.3356	0.8605	0.8593	0.3368	0.6890
Advanced Model	0.7882	0.4257	0.6013	0.8278	0.9073	0.4986	0.7916
Pre-trained models							
ResNet50	**0.8953**	**0.6748**	0.7757	**0.9206**	0.9508	0.7217	0.9313
VGG16	0.8674	0.5867	0.8224	0.8770	0.9588	0.6848	0.9170
ViT	0.8756	0.6165	0.7664	0.8988	0.9477	0.6833	0.9159
Ensemble (ResNet50+ViT)	0.8903	0.6429	**0.8411**	0.9008	**0.9639**	**0.7287**	**0.9347**

Within our handcrafted models, we observed clear performance gains with architectural complexity. Our Simple Model gave us a baseline performance for a single convolutional layer and our Deep Model gave us incremental performance gains through an additional number of convolutional layers. Our Regularized Deep Model gave us substantial gains in accuracy with L2 regularization, batch normalization and dropout at the cost of reduced recall. Our Advanced Model gave us the best performance among our handcrafted structures with the fusion of multimodal information from a separate pathway for tabular metadata.

Despite these advancements, the difference in performance between our best handcrafted model and pre-trained models is huge. ResNet50 is over 10 percentage points higher in accuracy (0.8953 vs. 0.7882) and 22 percentage points higher in F1-score (0.7217 vs. 0.4986). This indicates the massive advantage of transfer learning from models pre-trained on very large datasets such as ImageNet even when the target domain differs from the source domain.

Our results are in line with existing research in fracture detection, confirming the reality that DL techniques are promising in this endeavor. The performance measures are on par or slightly better than existing research with comparable architectures for fracture detection in other extremities. The effectiveness of ensemble techniques and ImageNet transfer learning in spite of domain difference between natural images and X-rays is also confirmed by our study.

6 Limitations and future work

6.1 Limitations

The major limitation that we faced in this project was the lack of available data. Pre-trained models such as ResNet50, ViT and VGG16 have been trained on massive datasets like ImageNet which has enabled them to learn highly generalized transferable patterns and as a result these models require comparatively far less data which is easily satisfied with the dataset we currently use in this project allowing them to achieve significantly much better performance as compared to handcrafted models.

On the contrary, the handcrafted models are designed from scratch and require huge amounts of data to learn meaningful and generalized patterns. With the current dataset size, increasing the model complexity beyond the advance model's structure leads to overfitting, restricting us from leveraging their full potential. If a larger dataset is made available, it would allow us to train and create far more complex models and the impact of considering additional (tabular) data could become more impactful, potentially bridging the gap and even surpassing the pre-trained models which do not inherently utilize such information and only use the x-ray image data.

6.2 Future work

In the future, this project could be extended in several directions. One such direction would be fracture localization, where the models detect and demarcate the fractured regions making the predictions more interpretable. Another extension is fracture characterization which not only provides us information on whether or not a fracture exists but also provides insights into the type / class of the fracture providing more diagnostic information. Another direction would be to consider multiple views and create a model that uses an appropriate architecture that could processes multiple images simultaneously and detect the existence of a fracture, potentially increasing the diagnostic performance. Since the csv file in our dataset

contains a variable `feature_count`, which we have ignored for our current setting we could also extend this project to not only predict the presence of fractures but also the number of fractures, providing additional diagnostic information.

7 Interpretability and ethics

7.1 Interpretability

There is an inherent trade-off between interpretability and model strength and although DL models delivers superior performance, it comes at the cost of reduced transparency. DL models' opaque nature presents strong barriers towards incorporating them into clinics, most notably in life-critical medical applications.

The current models do not have clear mechanisms for interpretability. Both, the simple architectures of the simple, deep, regularized deep and advanced models and the complex architectures of the pre-trained (ResNet50, ViT and VGG16) do not provide any reasoning during their decision-making process. Therefore, clinicians might show reluctance to trust systems that are still unexplainable or unverifiable.

Several methods can enhance the interpretability of our models. For instance, Class Activation Mapping (CAM), which includes Grad–CAM, has the potential to highlight areas of images that contribute significantly to classification results, thus helping radiologists verify the regions that the model believes to be relevant. Additionally, feature visualization methods can depict the features learned by various layers, thus providing insights into the hierarchical patterns that the models detects. Alternatively, example–based explanations can involve retrieving similar cases from the training data at the time of prediction, offering points of reference for clinicians to understand the reasoning behind model decisions. Lastly and on a simpler note, something we actually already provide that might help in gaining interpretability is uncertainty quantification (classification probability). Providing confidence scores or measures of uncertainty along with predictions, help clinicians identify cases where they need to be more cautious about the model's outputs

7.2 Ethics

The primary ethical concern is patient safety. While our models demonstrate promising performance, they are not perfect and may miss fractures (false negatives) or incorrectly identify normal bones as fractured (false positives). Both error types can lead to adverse outcomes, false negatives may result in delayed treatment, prolonged pain, improper healing and many more such complications, while false positives may lead to unnecessary additional imaging (with associated radiation exposure), specialist referrals, medical interventions, etc.

Models trained on limited or biased datasets may highlight and amplify existing healthcare disparities. For instance, our model was trained on a specific dataset that may not represent the full diversity of patient populations across age, sex, race, etc. Such disparities across demographic groups could lead to varied diagnostic results and may affect model performance in ways not captured by our evaluation metrics.

Therefore, these systems should be implemented as assistive tools for radiologists rather than autonomous diagnostic systems.

8 Conclusions

Our mini-project demonstrates the significant performance advantage of transfer learning for fracture detection in diverse anatomical regions. Pre-trained networks significantly outperformed hand-crafted architectures, with ResNet50 achieving 89.53% accuracy and 0.9313 ROC-AUC compared to 78.82% accuracy and 0.7916 ROC-AUC for our top-performing handcrafted model. These results confirm the high transferability of ImageNet learned features to medical imaging tasks in spite of inherent differences between natural and X-ray images. Ensemble averaging of ResNet50 and ViT provided the overall best performance (ROC-AUC of 0.9347), indicating that architectural diversity enables complementary feature detection to enhance diagnostic capability.

Our multimodal approach with tabular metadata (scanning angle, body part and hardware status) along with image features was the main contribution, outperforming image-only approaches by a significant margin among our handcrafted models demonstrating the value of contextual information in the context of fracture detection.

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A Original CSV variables

Variable	Type / Values	Description
`hand`, `leg`, `hip`, `shoulder`	Binary (0, 1)	Indicates which body part is visible in the X-ray. A value of 1 means that the corresponding body part is involved, while 0 means it is not.
`mixed`	Binary (0, 1)	If 0, exactly one of `hand`, `leg`, `hip`, or `shoulder` is marked as 1. If 1, more than one of these variables is 1, indicating that the image includes multiple body parts.
`fracture`	Binary (0, 1)	1 indicates that a fracture is present; 0 indicates that no fracture is present.
`fracture_count`	Integer (0, 1, 2, ...)	Provides the number of fractures detected in the image. A value of 0 means no fracture, while a positive integer indicates the number of fractures.
`frontal`, `lateral`, `oblique`	Binary (0, 1)	Indicates the scan angles used in the X-ray. A value of 1 means that the corresponding angle is present in the image.
`multiscan`	Binary (0, 1)	1 if more than one scan angle is taken (i.e., more than one of `frontal`, `lateral`, or `oblique` is 1); otherwise 0.
`hardware`	Binary (0, 1)	1 indicates that foreign hardware (such as screws, plates, etc.) is visible in the X-ray; 0 indicates it is not.

Table 2: Description of variables from the CSV file.

B Modified CSV variables

Body part code	Description
1	Shoulder
2	Hip
3	Hip and Shoulder
4	Leg
5	Leg and Shoulder
6	Leg and Hip
7	Leg, Hip and Shoulder
8	Hand
9	Hand and Shoulder
10	Hand and Hip
11	Hand, Hip and Shoulder
12	Hand and Leg
13	Hand, Leg and Shoulder
14	Hand, Leg and Hip
15	All (Hand, Leg, Hip and Shoulder)

Table 3: Mapping of the `body_part` variable.

Angle code	Description
1	Oblique
2	Lateral
3	Lateral and Oblique
4	Frontal
5	Frontal and Oblique
6	Frontal and Lateral
7	Frontal, Lateral and Oblique

Table 4: Mapping of the `angles` variable.